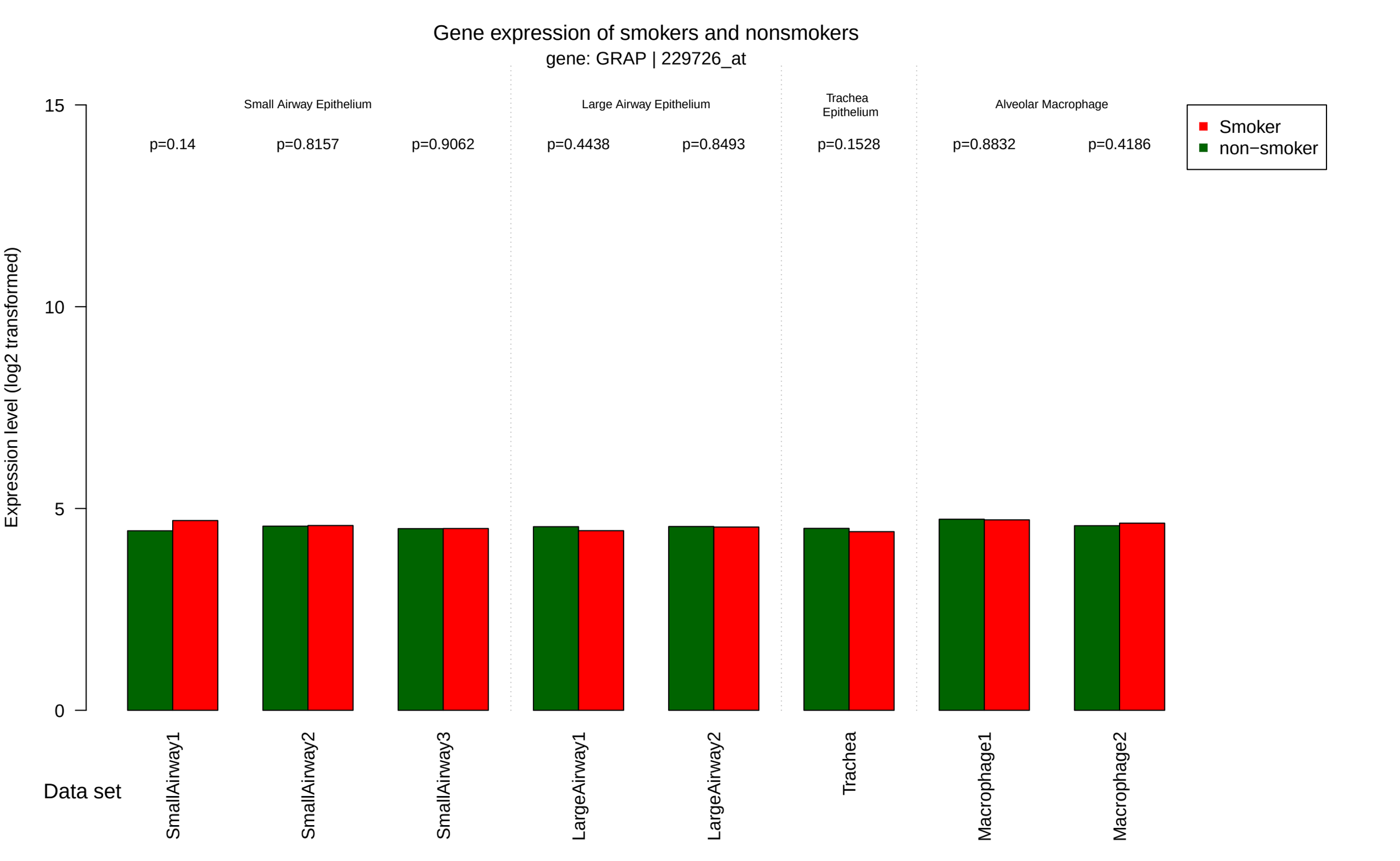
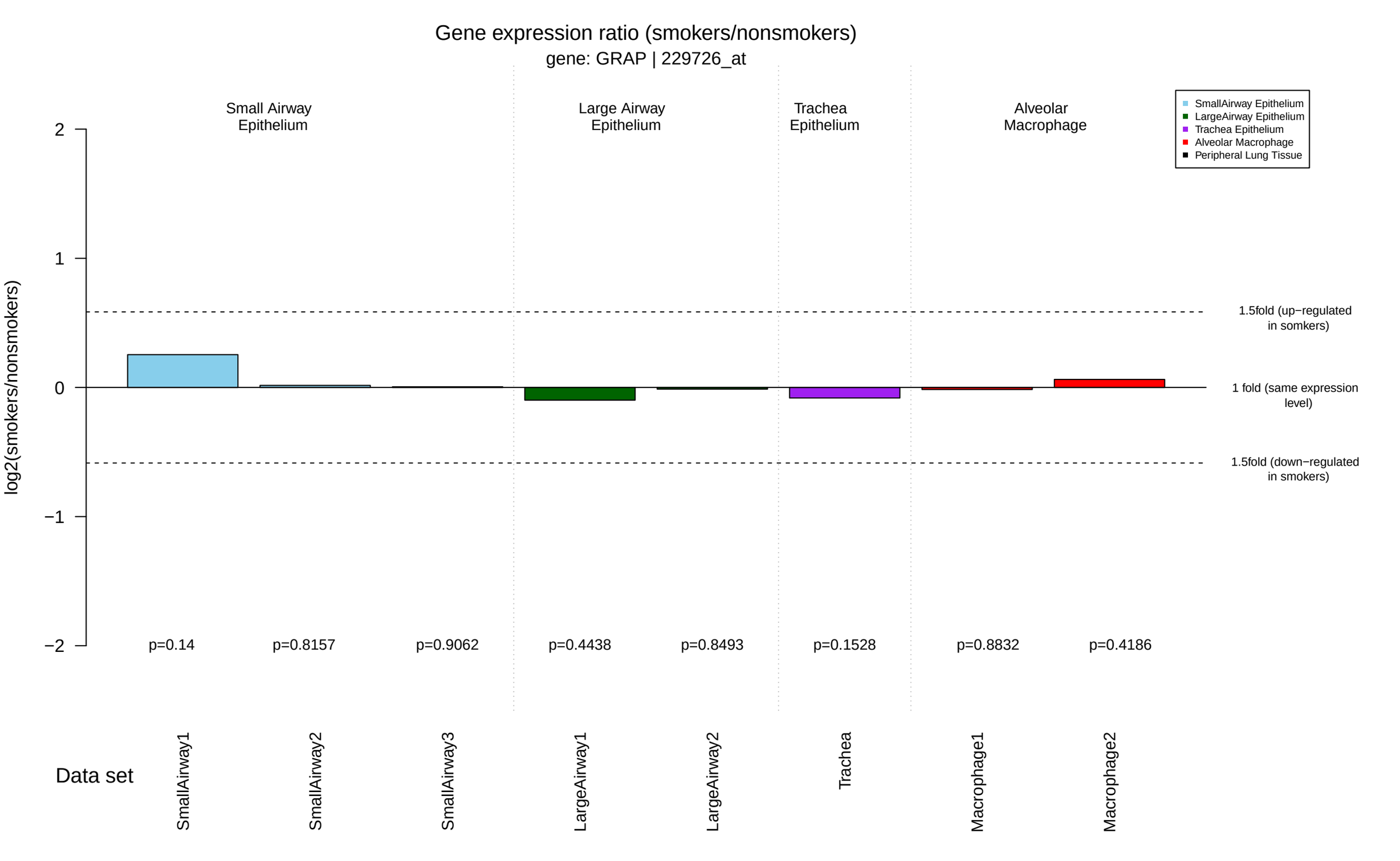
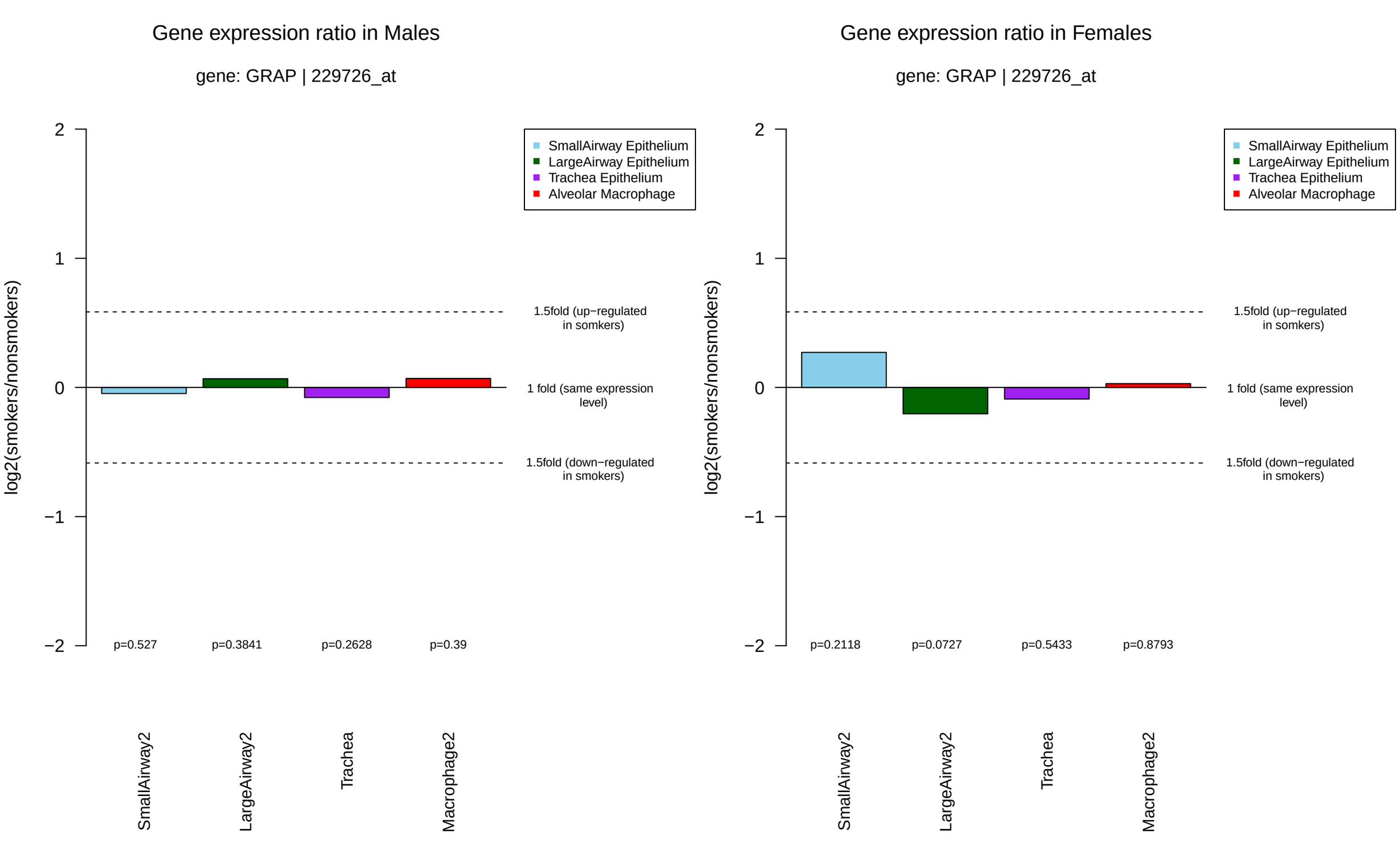
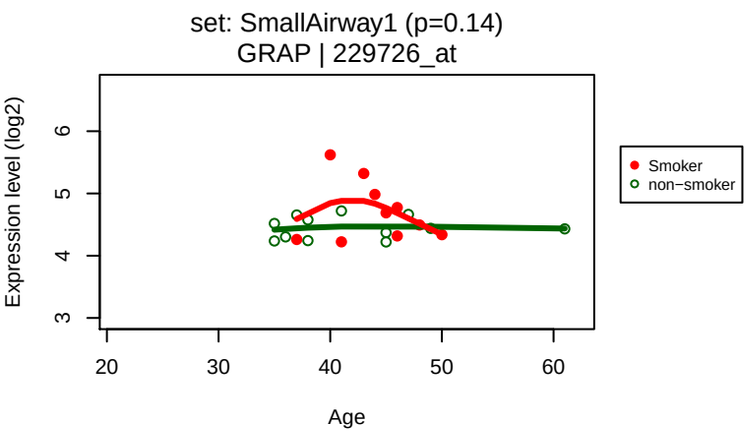
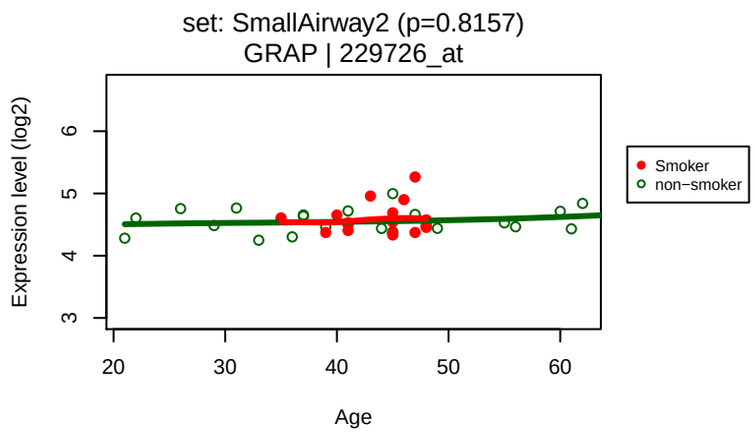
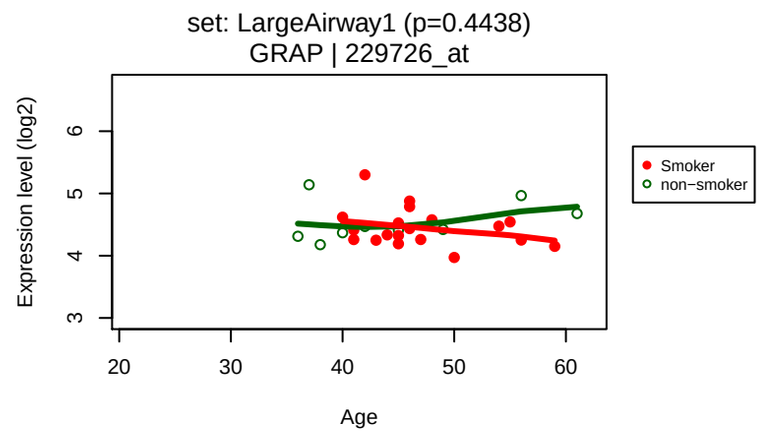

Gene expression of smokers and nonsmokers
gene: GRAP | 229726_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.14
p=0.8157
p=0.9062
p=0.4438
p=0.8493
p=0.1528
p=0.8832
p=0.4186
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

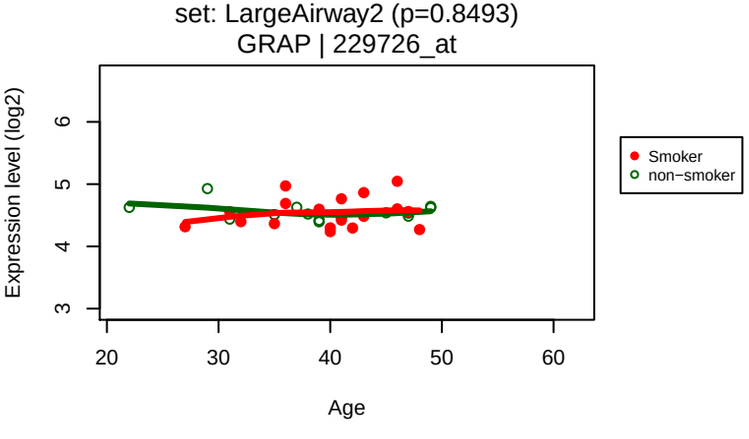

Gene expression ratio (smokers/nonsmokers)
gene: GRAP | 229726_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.14
p=0.8157
p=0.9062
p=0.4438
p=0.8493
p=0.1528
p=0.8832
p=0.4186
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

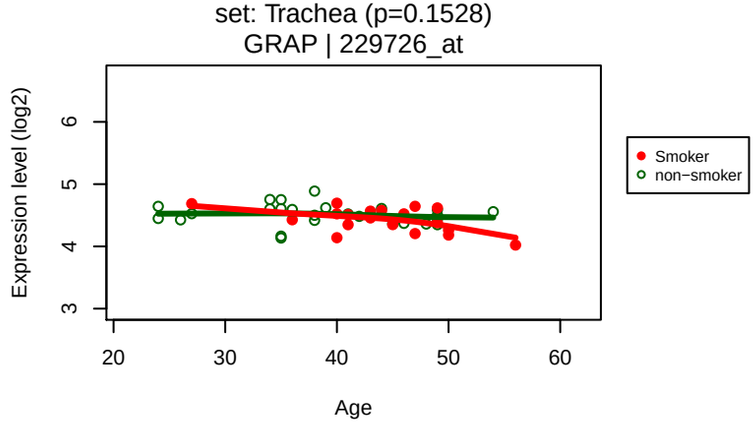

Gene expression ratio in Males
gene: GRAP | 229726_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.527
p=0.3841
p=0.2628
p=0.39
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRAP | 229726_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2118
p=0.0727
p=0.5433
p=0.8793
SmallAirway2
LargeAirway2
Trachea
Macrophage2

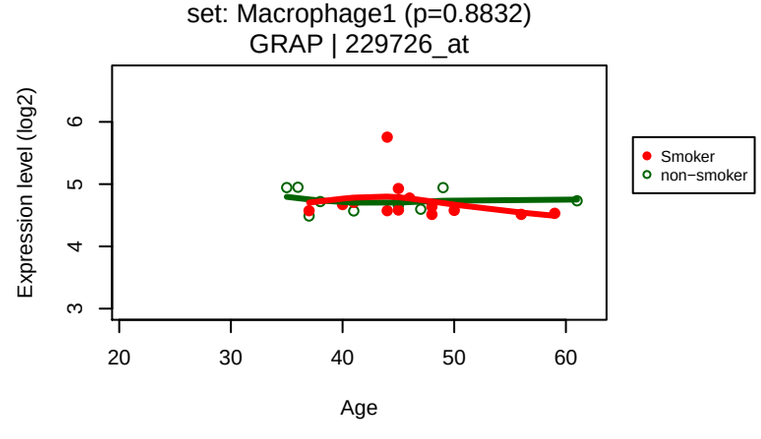

set: SmallAirway1 (p=0.14)
GRAP | 229726_at
Expression level (log2)
Age
Smoker
non−smoker

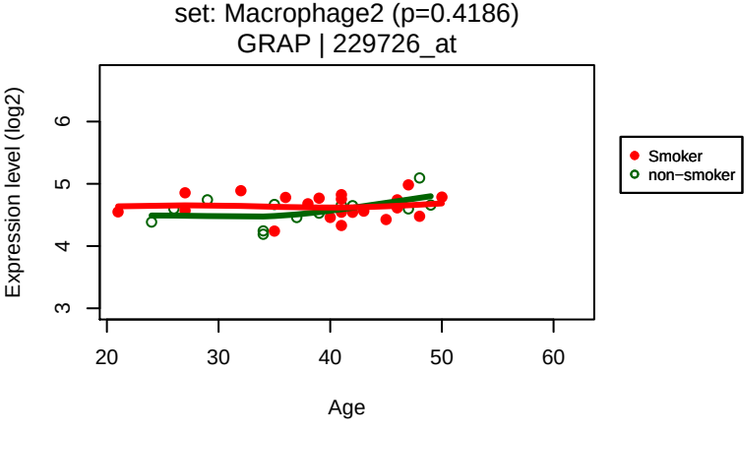

set: SmallAirway2 (p=0.8157)
GRAP | 229726_at
Expression level (log2)
Age
Smoker
non−smoker

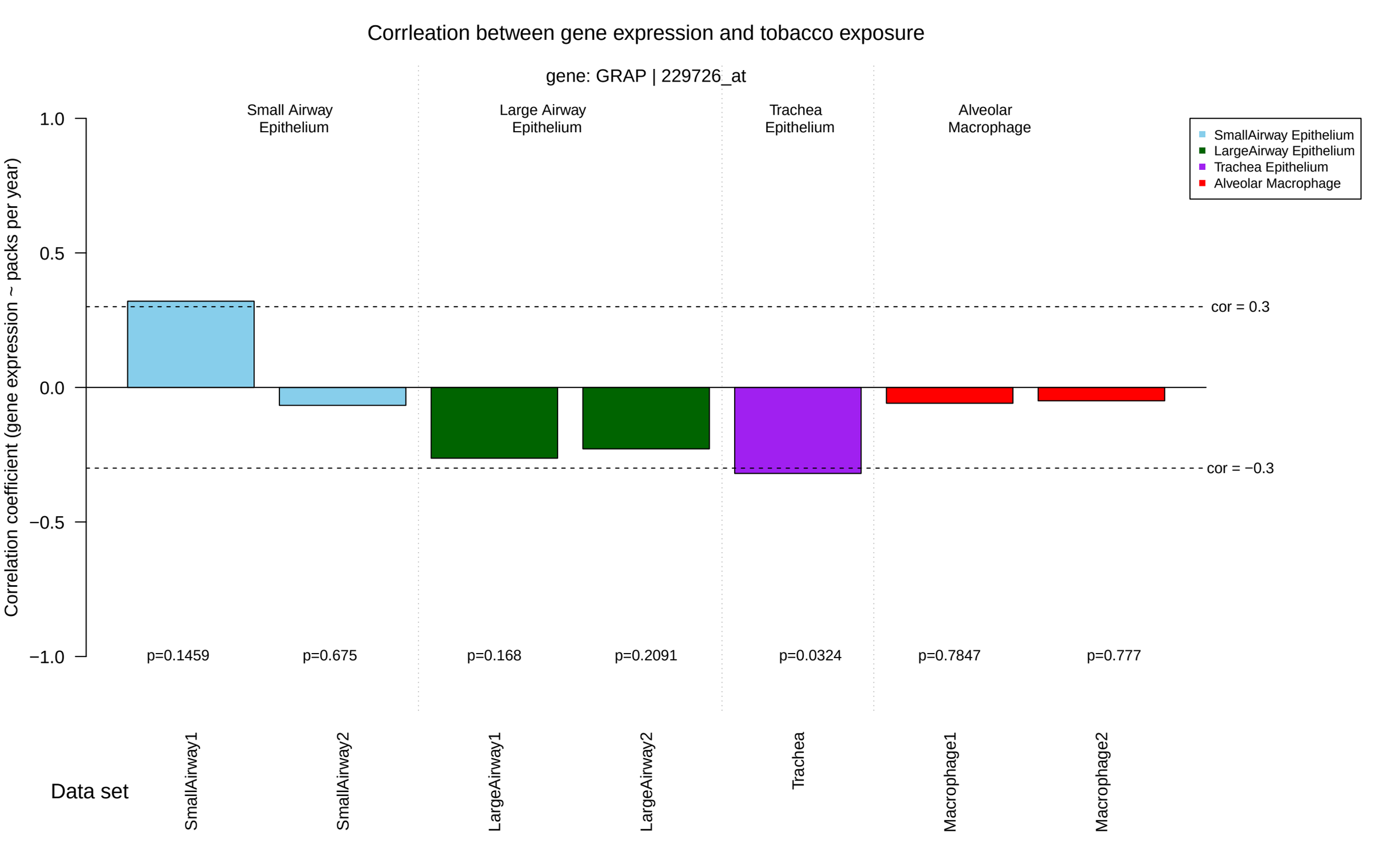

set: LargeAirway1 (p=0.4438)
GRAP | 229726_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8493)
GRAP | 229726_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1528)
GRAP | 229726_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8832)
GRAP | 229726_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4186)
GRAP | 229726_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRAP | 229726_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1459
p=0.675
p=0.168
p=0.2091
p=0.0324
p=0.7847
p=0.777
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2